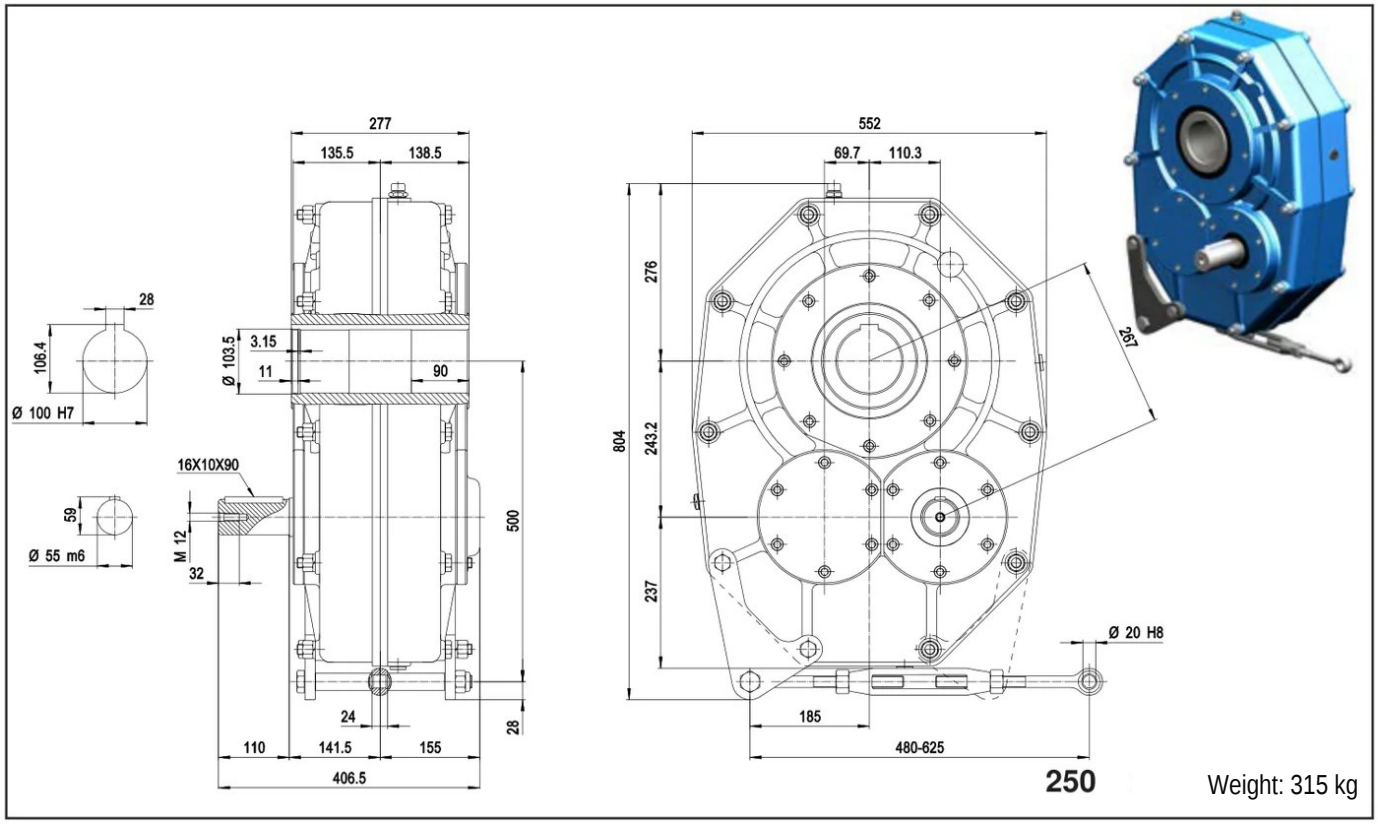
PRODUCT DATA SHEET

OCR RP 250

Shaft-mounted gear reducer | supplied and supported by Q² Machines

WEIGHT	RATED POWER	OUTPUT TORQUE	RATIOS
315 kg	**15–155 kW**	**4170–11280 Nm**	**5**
complete unit	across listed ratios	catalogue, service factor 1	single and double stage

DIMENSIONS (mm)



CONSTRUCTION

- Grey cast-iron housing with reinforcing and cooling ribs
- Case-hardened helical steel gears, controlled or ground tooth profile
- Ball and/or taper-roller bearings
- Hollow output shaft in Fe510 steel, with keyway and circlip groove
- Oil-bath lubrication with filler, level and drain plugs
- Double-lip seals, blue RAL 5010 finish
- Torque arm with brackets; backstop available on request

Ratings are catalogue values at service factor 1. Confirm selection for your duty with Q² Machines before ordering.

STAGE	RATIO	OUTPUT n2 (rpm)	INPUT n1 (rpm)	POWER (kW)	TORQUE (Nm)
Single	3.43	355	1218	155	4170
Single	3.43	280	960	132	4509
Single	3.43	220	755	112	4849
Single	3.43	180	617	98	5188
Single	3.43	140	480	81	5528
Single	3.43	110	377	67.6	5867
Single	3.43	90	309	58.5	6206
Double	9.87	140	1382	96	6547
Double	9.87	110	1086	83	7205
Double	9.87	90	888	74.1	7862
Double	9.87	71	701	63.3	8520
Double	9.87	56	553	53.8	9178
Double	9.87	45	444	46.3	9835
Double	9.87	35.5	350	39	10493
Double	15.5	90	1395	74	7853
Double	15.5	71	1101	62.6	8424
Double	15.5	56	868	52.8	8995
Double	15.5	45	698	45.1	9566
Double	15.5	35.5	550	37.7	10138
Double	15.5	28	434	31.4	10709
Double	15.5	22	341	26	11280
Double	19.2	71	1363	59	7939
Double	19.2	56	1075	48.9	8338
Double	19.2	45	864	41.2	8736
Double	19.2	35.5	682	34	9134
Double	19.2	28	538	28	9533
Double	19.2	22	422	22.9	9931
Double	19.2	18	346	19.5	10330
Double	24.5	56	1372	50	8525
Double	24.5	45	1103	41.5	8808
Double	24.5	35.5	870	33.8	9091
Double	24.5	28	686	27.5	9374
Double	24.5	22	539	22.3	9658
Double	24.5	18	441	18.7	9941
Double	24.5	14	343	15	10224

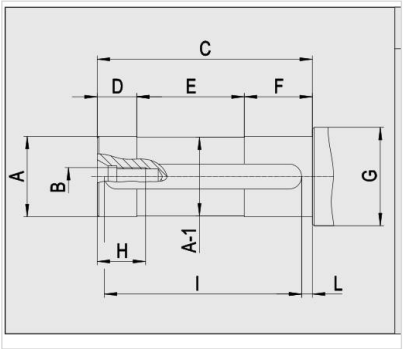
NOTES

- Rated power and torque are catalogue values for a service factor of 1 (OCR RP 2C catalogue, 2014).
- Maximum input speed 1400 rpm; preferred 710 rpm.
- Efficiency about 0.98 single stage, 0.96 double stage.
- Gear capacity calculated to ISO 6336, DIN 3990 and AGMA 2001-B88.
- Drawings in DXF, PDF and STEP formats, spares, and selection help are available from Q² Machines on request.

MOUNTING OPTIONS

Shaft-mounted: no coupling, foot or base plate needed

Installation on the machine shaft

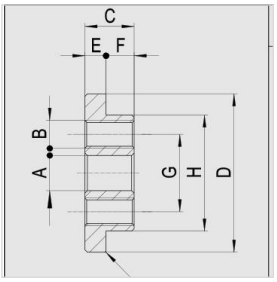


The reducer fits directly onto the shaft of the driven machine. Machine shaft tolerance h6, j6 or k6 as required; bore tolerance H7.

- Mount, fix and remove with tension rods and extractors, using the tapped hole in the end of the machine shaft and an installation washer.
- Clean and lubricate the contact surfaces before mounting to prevent rust.
- Do not force the reducer on with blows or jerks: this damages bearings and seals.
- Mount so the reducer is not subject to vibration.

Machine shaft, RP 250

ØA (mm)	B	C	D	E	F	ØG	H	I	L
100 h6/j6/k6	M20	248	61	97	90	115	50	220	14

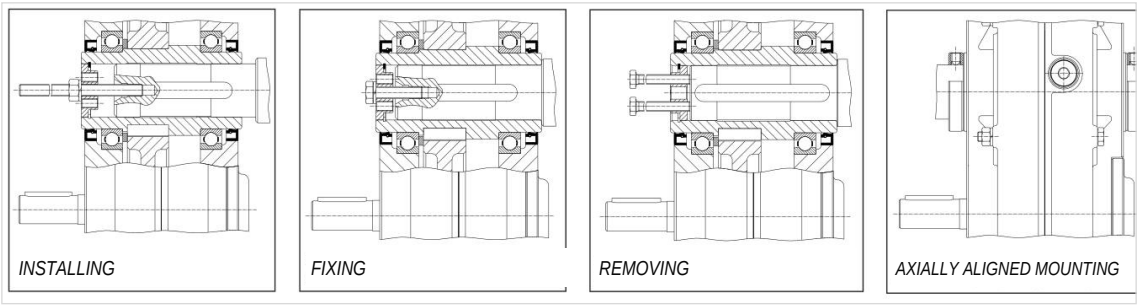


Installation washer, RP 250

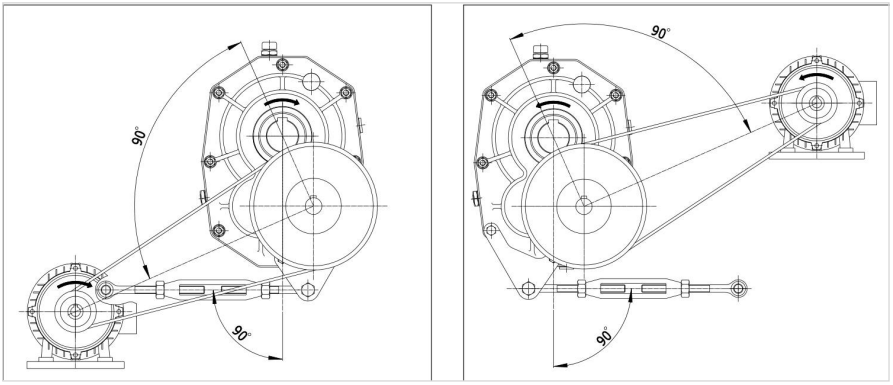
ØD h10	ØA	B	C	E	F	G	ØH
100	21 (M20)	M14	25	11	14	52	82

Make the washer from the dimensions shown; sharp edge on the outer rim. Bore tolerance H7.

Installing, fixing, removing and axially aligned mounting



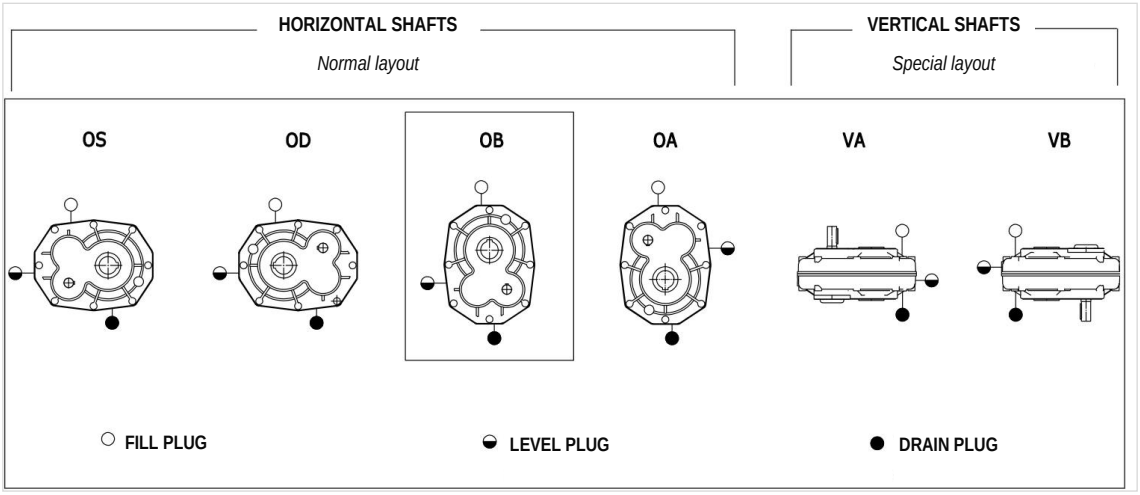
Torque arm arrangement



- Position the torque arm so it works in tension, at the angles shown. The arm also holds the belts at the right tension.
- For rotation in both directions or heavy duty, use two opposing torque arms.
- Fit torque limiters or safety devices where overloads are foreseen.

MOUNTING POSITIONS

Standard layout is OB; change position by swapping the plugs



Unless ordered otherwise, reducers are supplied in the OB layout. The other horizontal positions (OS, OD, OA) can be set by the customer by inverting the plugs. Vertical positions VA and VB are special and must be specified with the order.

LUBRICATION RECOMMENDATIONS

Supplied without oil

RP Series reducers are oil bath and splash lubricated and are supplied WITHOUT OIL. Use ISO VG mineral oil according to the ambient temperature. Fill to the level plug and avoid overfilling. When input speed is below 355 rpm, or in vertical positions, the upper bearings are permanently greased.

AMBIENT TEMPERATURE	VISCOSITY ISO VG
Below -10 °C	**100**
-10 to +10 °C	**150**
+10 to +30 °C	**220**
Above +30 °C	**320**

EXAMPLES OF SUITABLE OILS

AGIP Blasia
ARAL Degol BG
BP Energol GR-XP
CASTROL Alpha SP
ESSO Spartan EP
MOBIL Mobilgear Serie 6
SHELL Omala Oil
TEXACO Meropa
TOTAL Carter EP

Oil quantity for RP 250, litres, double reduction (RP2C)

OS	OD	OB (standard)	OA	VA	VB
14.1	11.7	9.7	19	11.3	18

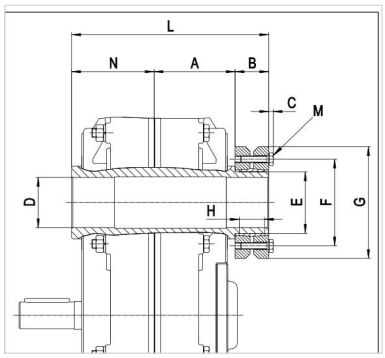
- For positions between those in the table, interpolate. For single reduction units (RP1C, ratio 4), add 10%.
- Change the oil after a running-in period of 500 operating hours, check the level every 1000 hours and change the oil every 4000 hours.
- If oil temperature exceeds 80 °C, or under overloads, change the oil more often.

HANDLING, STORAGE AND SAFETY

- Allow space around the reducer for inspection and maintenance and enough air flow for cooling. Take care when lifting.
- Protect from weather and direct sun where possible. Store in a dry, clean place; every six months turn the gears a few times and check the grease in the seals and the protective coating on machined parts. Check more often in damp, corrosive or hot-cold environments.
- Fit protection where an accidental oil leak could cause serious damage, and guards over the belt drive.
- Supplied painted blue RAL 5010; shaft ends and seals are protected with water-repellent grease.

OPTIONS

Available on request; confirm availability with Q² Machines



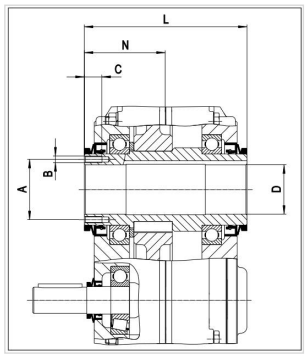
Shrink disc (keyless locking)

A keyless device to mount and fix the machine shaft to the reducer, removing the key and installation washer and giving positive control of the torque transmitted. A special hollow shaft is needed, with the dimensions below.

RP 250, mm unless stated

ØD H7	L	N	A	B	C	Screws	ØE h8
100	357	151	146	60	8	M12	140

F	ØG	H	Type	Screw torque Nm	Torque Nm	Axial force N	kg
175	230	46	d140	100	17,600	352,000	10

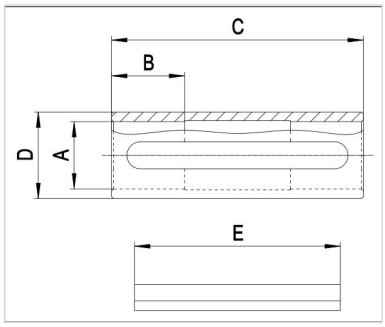


Special seals

For dusty or aggressive environments where regular seal maintenance is impractical. An additional rotating front seal (V-ring on a wear plate) runs with the shaft against a fixed surface. A special hollow shaft is needed.

RP 250, mm

ØD H7	L	N	A	B	C	V-ring input	V-ring output
100	302	151	125	M12	32	60	150

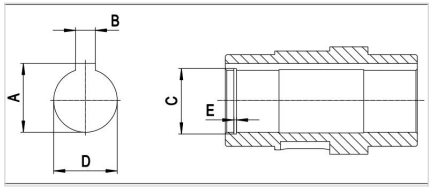


Reduction bushing with key

Lets you stock reducers with the maximum bore and fit a smaller shaft diameter with a bushing, which cuts stock. The bushing is axially locked with spacers and installation washers and is 5 mm shorter than the hollow shaft. Applies to reducers supplied with the standard hollow shaft diameters.

RP 250, mm

ØD H7 (outer)	ØA H7 (bore options)	C	B	E
110	80, 90, 100	272	90	234



Hollow shaft: other bore diameters

Where possible, mount on a shaft the same size as the catalogue bore. Otherwise hollow shafts with other bore diameters can be supplied, up to the maximum bore. Imperial and American bore sizes can also be made, within the maximum.

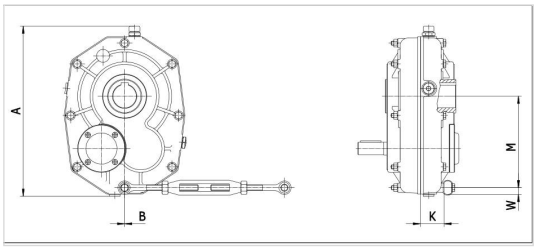
RP 250, Ø mm H7

Catalogue bore	Maximum bore	Special bores available
100	110	85

AVAILABLE HARDWARE

Torque arms, motor mount and backstop

The torque arm anchors the reducer and holds the belts at the correct tension. The standard supply is a torque arm with brackets (dimensions on page 1). Alternatives and accessories for RP 250 are below.

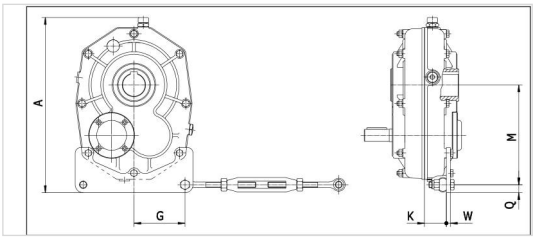


Torque arm without brackets

Supplied without the mounting brackets, for fixing to your own structure.

RP 250, mm

A	B	M	W	K
756	94.2	449	21	104

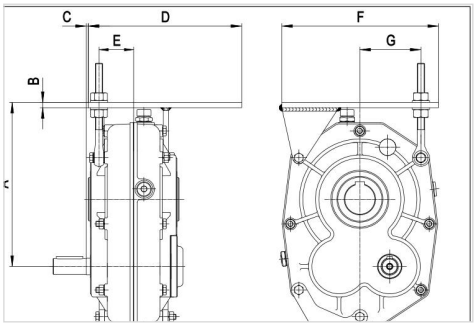


Torque arm with plate

Torque arm supplied with a mounting plate.

RP 250, mm

A	G	M	Q	W	K
752	190	450	28	15	101

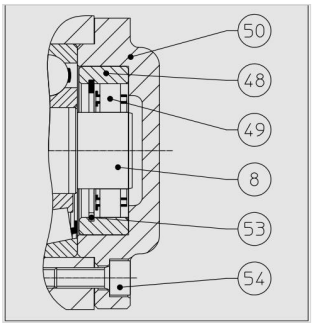


Motor mount

A compact motor-and-reducer assembly: the belt tension is adjusted at the motor mount torque arm. One motor mount is available for each size, with the maximum motor frame shown.

RP 250, mm

Max. motor (B3)	A	B	C	D	E	F	G	Weight kg
225 M	545	16	3	505	116	555	228.6	41.7



Backstop (anti roll-back device)

A freewheel with locking elements, fitted on the input shaft opposite the shaft end, which stops the output shaft turning the wrong way. It can be fitted without removing the reducer from the machine shaft. State the output shaft direction of rotation when ordering, viewed from the side opposite the input shaft. Not available with a double-extended input shaft; on sizes 85 to 165 not with an opposite-side input shaft.

RP 250

Rated input torque M1	Max. overload (1.7 x M1)
640 Nm	1088 Nm

Check the backstop load capacity: the worst case is low ratios, low service factors and low speeds.

APPLICATIONS

Where shaft-mounted reducers work hardest

Shaft-mounted reducers drive conveyors, screw conveyors, elevators, mixers and crushers, and are widely used on feeders, screens, stackers and material washers. They suit applications where a belt drive between the motor and the reducer is practical.

- Quick and easy to install: no coupling or base plate needed.
- Rotation in one direction can be prevented with a backstop.
- Output speed can be adjusted by changing the motor and reducer pulleys as well as the gear ratio.
- A range of special executions and accessories.









OTHER SPECIAL EXECUTIONS

- Special reduction ratios 6.3, 12.5 and 28 on request (not on every size).
- Double-extended input shaft, or input shaft on the opposite side to the catalogue layout.
- Anti-fretting and anti-wear treatment of the hollow shaft.
- Vertical mounting positions VA and VB.

ORDERING

To select and order, send Q² Machines the required power or torque, output speed, duty and operating position, backstop and torque arm requirements, and the machine shaft diameter. Quote the catalogue designation, for example RP2C150-19.5 ARO with torque arm with brackets. Dimensional drawings in DXF, PDF and STEP formats and spare parts are available on request.